Ishaan Pathak

Southfield, Michigan (Open to relocate) | ishaan175pathak@gmail.com | LinkedIn | Github| Researchgate | Portfolio

Education

Lawrence Technological University — Southfield, MI
Master's of Science in Computer Science — *August 2024 – July 2026*
Teaching Assistant: Python & Robotics

Devi Ahilya VishwaVidyalaya — Indore, India
Bachelor of Computer Application (Honours) — *August 2019 – July 2023*

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML, CSS
AI/ML: PyTorch, TensorFlow, Keras, Transformers (Hugging Face), Computer Vision, RNNs, CNNs, YOLO, scikit-learn, scipy
Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Semantic Search
Web Development: React.js, Node.js, Express.js, Flask, Django, Spring Boot, Three.js, REST APIs
Cloud & DevOps: AWS (Lambda, DynamoDB, S3 Buckets), Docker, Kubernetes, CI/CD, GitHub Actions
Databases: MySQL, PostgreSQL, MongoDB, Oracle, Redis

Experience

Research Assistant — February 2025 – July 2026
Lawrence Technological University — *Southfield, MI*

- **Achieved 81%** mAP on YOLO-based crack detection models by curating and **annotating a 10,000+** image dataset
- **Developed** state-of-the-art Computer Vision models for medical detection, contributing to automated medical research.
- **Assisted** professor in **reviewing, refining, publishing** research manuscripts for clarity, rigor, and publication readiness.

Student Web Developer — May 2025 – August 2025
Lawrence Technological University — *Southfield, MI*

- **Developed** Web app for anonymous essay submissions, enabling over **50+ students** to participate in bias-free evaluations.
- **Integrated** a Gen AI model to benchmark automated grading against human assessments, **improving** consistency by **25%.**
- **Optimized** the grading workflow, **reducing** manual evaluation time by **40%** and grading throughput **increased by 35%**.

AI/ML & Data Engineer — July 2023 – July 2024
Essar Agro — *Indore, India*

- **Standardized 1M+** procurement records, reducing data preparation time by **10%** through a Python ingestion pipeline.
- **Automated** livestock medication dosing with **85% accuracy** by training Keras LSTM/GRU models on client text data.
- **Reduced** deployment time by **30%** by building Docker and GitHub Actions CI/CD workflows for ML pipelines.

Junior Machine Learning Engineer — June 2022 – February 2023
GenieTalk.ai — *Indore, India*

- **Developed** and **optimized** machine learning models for text classification using TensorFlow and scikit-learn.
- **Enhanced** algorithm performance and **integrated** AWS for scalable model deployment.
- **Developed** Application Programming Interface (APIs) for model integration, enhancing response times by **20%.**

Projects

Building Segmentation & Geospatial Extraction | *PyTorch, U-Net, SegFormer, OpenCV, GeoJSON*

- **Achieved 94.20% Dice** and **89.03% IoU** on WHU building segmentation by ensembling U-Net and SegFormer-B0.
- **Automated** mask-to-polygon conversion for geospatial analysis via an end-to-end inference pipeline with GeoJSON output.
- **Reached 0.86 F1** at 0.5 IoU for building-level extraction by benchmarking CNN, transformer, and ensemble models.

Social Media Analysis | *Apache Spark, Scikit-Learn, Python (Matplotlib, Seaborn, Pandas), SQL*

- **Engineered** an **Apache Spark** pipeline to process high-volume tweet data using distributed computing.
- **Implemented K-Means** and **PCA** to automate content segmentation and identify sentiment trends.
- **Automated** visual reporting via **Matplotlib/Seaborn** to deliver actionable insights for brand monitoring.

Personalized Search Reranker | *BERT, Cross-Encoders, TREC DL, PyTorch*

- **Boosted** search accuracy (**MRR@10 to 0.0406**) by developing a reranking pipeline integrated with clickstream data.
- Predicted **query-document relevance** by fine-tuning Sentence-BERT on MS MARCO using binary cross-entropy.
- Enhanced **inference efficiency by 15%** by scaling Python preprocessing and optimizing **AdamW** training.

Research, Awards & Certifications

- Publications: **AI-Powered Music Generation from Sequential Motion Signals** (IJIIS 2025); **Automated Brain Tumor Detection via YOLOv11** (JCTR 2025)
- Research: **Smart Infrastructure Damage Detection** using Computer Vision for structural health monitoring.
- **Outstanding Research Assistant in Data Science Award** — Lawrence Technological University